INCH-POUND

MIL-C-48139B (AR)
15 March 1990
SUPERSEDING
MIL-C-48139A (AR)
17 May 1978

MILITARY SPECIFICATION

CARTRIDGE, 105MM, TPDS-T, M724A1 LOADING, ASSEMBLING AND PACKING

This Specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing of the M724A1, 105mm, TPDS-T cartridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg.4D, 700 Robbins Avenue, Philadelphia, PA 19111-5904.)

2.1.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9278417 - Packing and Marking for Container, Ammo, Fiber, PA72 for Cartridge 105MM, TPDS-T, for Gun M68
9278500 - Cartridge, 105MM, TPDS-T, M724A1, Assembly and Marking Diagram
9278419 - Packing and Marking for Box, Packing, for Container, Ammo, Fiber, PA72 for Cartridge, 105MM, TPDS-T, for Gun M68
8828098 - Gage, Profile and Alignment
9341656 - Packing and Marking for Box, Wirebound, Packing Ammunition, for Cartridge 105MM, TPDS-T, M724, M724E1 and M724A1 for Gun M68

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with specifications and drawings.

3.2 Cartridge. The cartridge shall comply with all requirements on drawing 9278500 and with all requirements in applicable specifications.

3.3 Propellant weight. The weight of the propellant shall not differ by more than (±) 0.12 ounce avoirdupois from that designated by the loading authorization for the particular lot of propellant used.

3.4 Cartridge case crimp (bullet pull). The cartridge case shall separate from the shot at a load between 1300 pounds and 4000 pounds.

3.5 Security of sub-projectile. The sub-projectile shall not rotate in the projectile sabot when subjected to 50 ± 5 inch-pounds torque in a counterclockwise direction when facing the nose of the projectile.

3.6 Rotating band looseness and seating. Within one hour of assembly with the cartridge case, the rotating band shall be seated against the sabot and shall not rotate on the projectile assembly when subjected to manually applied torque in a clockwise direction, as viewed from the rear of the projectile.

3.7 Ballistics tests. The cartridge shall comply with the following requirements.

3.7.1 Muzzle velocity. The corrected muzzle velocity of the cartridge, when fired at 70°F, shall not be more than 5105 feet per second (fps) nor less than 4995 feet per second (fps).

3.7.2 Target impact dispersion. The maximum probable-error shall be 0.25 mils horizontal and 0.22 mils vertical (see 4.4.3.6.3).

3.7.3 Tracer functioning. The tracer shall ignite and continue burning for 1000 meters.

3.7.4 Safety. There shall be no hangfires or indications of metal parts failure in the gun bore or in flight, prior to impact, except as will normally occur with respect to the discarding of the sabot.

3.7.5 Flight. There shall be no evidence of erratic flight greater than that specified in 4.4.3.6.6. (See 6.2).

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3.7.6 Misfire. There shall be no misfires during any of the tests conducted.

3.8 Workmanship.

3.8.1 General. The cartridge and all components shall be free of cracks or burrs, chips, dirt, grease, rust or other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. The cartridge and packaging shall not be damaged in any manner during any loading, assembling or packaging operation.

3.8.2 Painting and marking. The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such paintings and markings have been marred or obliterated. Bare spots in painting, other than marking, are allowable provided that they total not more than 1/4 square inch, and that the maximum dimension of any individual bare spot does not exceed 3/16 inch.

3.9 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Submission of product. At the time each complete lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection (See 4.3)

b. Quality conformance inspection (See 4.4).

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4.3 First article inspection.

4.3.1 Submission. At the beginning of initial production, ten (10) cartridges shall be subjected to the bullet pull test and ninety (90) cartridges shall be inspected in accordance with this specification and forwarded to the proving ground for ballistic testing (see Table I). The sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production, and shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by the procuring agency shall be submitted whenever there is a lapse in production for a period in excess of ninety days or whenever a change occurs in materials, equipment, processes, procedures, drawings, specifications, or sources of supply such as to significantly affect product uniformity as determined by the responsible technical agency.

4.3.2 Test plans. The sample shall be subjected to the functional tests as specified in Table II. The first article test shall consist of all phases listed in Table I.

4.3.3 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.4 Rejection. See MIL-A-48078.

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Table I First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48139B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
	Cartridge, 105mm, TPDS-T, M724A1				9278500
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination of defects	90			4.4.2.1 thru 4.4.2.4
	Bullet pull test	10		3.4	4.5.1
	Ballistic tests				
	Phase 1 (70°F - New tube, see 6.11)	15			
	Muzzle velocity			3.7.1	4.5.6.1.2
	Target-Impact-Dispersion			3.7.2	4.5.6.1.2
	Tracer functioning			3.7.3	4.5.6.1.1.8
	Safety			3.7.4	4.5.6.1.2
	Flight			3.7.5	4.5.6.1.1.9
	Spin rate		Info only		4.5.6.1.2
	Misfire			3.7.6	4.5.6.1.2
	Phase 2 (70°F - Worn tube, see 6.11)	15			
	Tracer functioning			3.7.3	4.5.6.1.1.8
	Safety			3.7.4	4.5.6.1.3
	Flight			3.7.5	4.5.6.1.1.9
	Muzzle velocity		Info only		4.5.6.1.3
	Target-Impact-Dispersion		Info only		4.5.6.1.3
	Spin rate		Info only		4.5.6.1.3
	Misfire			3.7.6	4.5.6.1.3

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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Table I First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
	Cartridge, 105mm, TPDS-T, M724A1				9278500
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Phase 3 (125°F - mid-life tube see 6.11)	15			
	Tracer functioning			3.7.3	4.5.6.1.1.8
	Safety			3.7.4	4.5.6.1.3
	Flight			3.7.5	4.5.6.1.1.9
	Muzzle velocity		Info only		4.5.6.1.3
	Target-Impact-Dispersion		Info only		4.5.6.1.3
	Spin rate		Info only		4.5.6.1.3
	Misfire			3.7.6	4.5.6.1.3
	Phase 4 (125°F - worn tube see 6.11)	15			
	Tracer functioning			3.7.3	4.5.6.1.1.8
	Safety			3.7.4	4.5.6.1.3
	Flight			3.7.5	4.5.6.1.1.9
	Muzzle velocity		Info only		4.5.6.1.3
	Target-Impact-Dispersion		Info only		4.5.6.1.3
	Spin rate		Info only		4.5.6.1.3
	Misfire			3.7.6	4.5.6.1.3

NOTES:

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Table I First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48139B (AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
	Cartridge, 105mm, TPDS-T, M724A1		3	3	9278500
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Phase 5 (-25°F - mid-life tube see 6.11)	15			
	Tracer functioning			3.7.3	4.5.6.1.1.8
	Safety			3.7.4	4.5.6.1.3
	Flight			3.7.5	4.5.6.1.1.9
	Muzzle velocity		Info only		4.5.6.1.3
	Target-Impact-Dispersion		Info only		4.5.6.1.3
	Spin rate		Info only		4.5.6.1.3
	Misfire			3.7.6	4.5.6.1.3
	Phase 6 (-25°F - worn tube see 6.11)	15			
	Tracer functioning			3.7.3	4.5.6.1.1.8
	Safety			3.7.4	4.5.6.1.3
	Flight			3.7.5	4.5.6.1.1.9
	Muzzle velocity		Info only		4.5.6.1.3
	Target-Impact-Dispersion		Info only		4.5.6.1.3
	Spin rate		Info only		4.5.6.1.3
	Misfire			3.7.6	4.5.6.1.3

NOTES:

AMSMC Form 1570, 1 Feb 85

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, one lot of cartridges shall contain:

a. Cartridge cases from not more than three lots used from any group of five consecutive lots from one manufacturer.

b. Projectile metal parts from not more than two consecutive lots of one lot interfix number from one manufacturer.

c. Propellant from one lot from one manufacturer.

d. Loaded primers of one lot interfix number from one manufacturer.

e. Loaded tracers of one lot interfix number from one manufacturer.

f. Base plug assemblies of one lot interfix number from one manufacturer.

4.4.2 Examination. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-A-48078 and MIL-STD-105, Inspection Level II. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government.

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Table II

Ballistic Test Summary

First Article - Initial Production - Regular Production

FA	IP	RP	°F Temp	Tube Wear	No. Rds	Acceptance Test MV	TID	Tracer	Safety	Flight	Misfire	Spin	Info Test MV	TID	Spin
xxx	xxx	xxx	70	New	15	xxx	xxx	xxx	xxx	xxx	xxx	---	---	---	xxx
xxx	xxx	---	70	Worn	15	---	---	xxx	xxx	xxx	xxx	---	xxx	xxx	xxx
xxx	---	---	125	Mid-Life	15	---	---	xxx	xxx	xxx	xxx	---	xxx	xxx	xxx
xxx	---	---	125	Worn	15	---	---	xxx	xxx	xxx	xxx	---	xxx	xxx	xxx
xxx	---	---	-25	Mid-Life	15	---	---	xxx	xxx	xxx	xxx	---	xxx	xxx	xxx
xxx	---	---	-25	Worn	15	---	---	xxx	xxx	xxx	xxx	---	xxx	xxx	xxx

FA - First article
IP - Initial production (three consecutive lots shall be accepted with respect to flight and safety. Thereafter, regular production acceptances shall start.)
RP - Regular production

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48139B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Primer, Prior to Assembly to Cartridge Case		8839499
			NEXT HIGHER ASSEMBLY 9278500

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Any flash hole missing		100%	3.2	Visual
2	Plug missing		100%	3.2	Visual
Major					
101	O-ring missing		0.40%	3.2	Visual
102	Deterioration of primer (if applicable)		Note	3.2	4.4.3.1
Minor	None defined				

NOTES:
1. See primer specification.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48139B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Case, Cartridge, After Assembly of Primer but Prior to Loading of Propellant	1 OF 1	9278500
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Primer stake missing or improper		100%	3.2	Visual
102	Primer bent or damaged		0.40%	3.2	Visual
103	Primer fails to withstand minimum disassembly torque		0.40%	3.2	Gage
104	Electrical resistance between primer electrode and cartridge case		100%	3.2	Gage
Minor					
201	Foreign matter in interior of case		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48139B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Shot Loading Assembly, Prior to Assembly to Cartridge Case	1 OF 1	9278499
			NEXT HIGHER ASSEMBLY 9278500

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Rotating band loose		100%	3.2	4.5.4
102	Plug and disc assembly fails to withstand reapplication of assembly torque while cement is still wet with no discernible movement		0.40%	3.1	Torque test
103	Obturator damaged or improperly assembled		0.40%	3.8	Visual
104	Excessive cement on plug and disc assembly		0.40%	3.8	Visual
105	Base plug assembly exceeds max. insertion depth under qualifying load	(a)	0.40%	3.1	Visual/Manual
106	Plug not fully seated (180°)		0.40%	3.1	Gage
Minor					
201	Marking misleading or unidentifiable including incorrect color		0.65%	3.8	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: (a) Apply a qualifying load of 450 ± 50 pounds to the base plug assembly. The plug shall display resistance to assembly and shall not enter the base further than .25 inches.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48139B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Cartridge Assembly		9278500
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety	(a)		3.7.4	4.4.3.6.5
Major					
101	Height from front face of cartridge case flange to primer electrode, min.		0.40%	3.2	Gage
102	Assembly damaged		0.40%	3.8	Visual
103	Profile and alignment		4.4.3.5	3.2	4.5.5
104	Cartridge case crimp		4.4.3.3	3.4	4.5.1
105	Subcaliber projectile security		100%	3.5	4.5.3
106	Ballistic test		4.4.3.6	3.7	4.5.6
Minor					
201	Exterior protective coating with bare spot		0.65%	3.8.2	Visual
202	Marking misleading or unidentifiable including incorrect color code		0.65%	3.8.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: (a) Rounds fired in ballistic test shall be observed for compliance with safety requirements.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48139B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Unsealed Fiber Container	1 OF 1	9278416
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Packing component missing		0.40%	3.8	Visual
Minor					
201	Container assembled improperly		0.65%	3.8	Visual
202	Glue or asphalt on cartridge		0.65%	3.8	Visual
203	Cartridge cannot be removed by hand		0.65%	3.8	Manual
204	Container cap cannot be removed by hand		0.65%	3.8	Manual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48139B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Sealed Fiber Container	1 OF 1	9278417
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tape incomplete or badly wrinkled		0.40%	3.8	Visual
102	Container cut or damaged through all impregnated layers		0.40%	3.8	Visual-Manual
103	Metal end loose or distorted		0.40%	3.8	Visual-Manual
Minor					
201	Tear tab length, min.		0.65%	3.1	Gage
202	Gap exceeding 1/8 inch between cap and body of container		0.65%	3.1	Visual
203	Marking misleading or unidentifiable		0.65%	3.8	Visual
204	Cuts, scuffs or gouges in outer layers		0.65%	3.8	Visual
205	Contents loose		0.65%	3.8	Manual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48139B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Sealed Wood Packing Box		9278419, 9341656
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged, exposing contents		0.40%	3.8	Visual
102	Strapping missing or loose		0.40%	3.8	Visual
103	Hardware missing, broken or incorrect		0.40%	3.8	Visual
104	D.O.D. symbol missing or incorrect		0.40%	3.8	Visual
105	Board loose or broken		0.40%	3.8	Visual-Manual
Minor					
201	Car seal missing, unsealed or improperly positioned		0.65%	3.8	Visual
202	Marking misleading or unidentifiable		0.65%	3.8	Visual
203	Hardware improperly engaged		0.65%	3.8	Visual-Manual
204	Strapping improperly assembled		0.65%	3.8	Visual-Manual
205	Handle missing or insecure		0.65%	3.8	Visual-Manual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Testing.

4.4.3.1 Check test for possible deterioration of primers. If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall satisfactorily pass the check test for deterioration specified in the primer specification immediately before the primers are assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the cartridge (see 6.4).

4.4.3.2 Propellant weight. (See 3.3) Propellant shall be weighed and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing and, if performed manually, shall be performed by another operator. Any propellant charge which fails to comply with requirements shall be classed defective and removed from the lot.

4.4.3.3 Cartridge case crimp (bullet pull). (See 3.4, 4.5.1 and drawing 9278500.) An alternate acceptance method shall be in accordance with a government approved generic Statistical Process Control plan. The sampling plan and procedure shall be three (minimum) samples at the start of production (after normal shut down between shifts), and three (minimum) consecutive samples thereafter from each crimping machine in operation for each two hour production. Acceptance criteria shall be that the process remain in control with a Process Performance (CpK) index equal to or greater than 1.33 for that portion represented by the sample with zero failures since last acceptance test.

4.4.3.4 Projectile weight. Fifty loaded projectiles per lot shall be randomly selected and weighed. The mean and standard deviation of the sample shall be recorded on the lot data card. Individual results shall be retained for future reference.

4.4.3.5 Profile and alignment. Profile and alignment of each crimped cartridge shall be verified in accordance with the following inspection plan:

a. Inspect 50 cartridges in 4 angular orientations approximately 90 degrees apart.

b. After 50 consecutive cartridges successfully chambergage, reduce inspection to 1 out of 100 in 4 angular orientations and all others in 1 random orientation.

c. Any failure will result in returning to step (a).

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Any cartridge which fails to be inserted or seated with the required force and velocity in any orientation and in accordance with the above sampling plan shall be rejected. When applicable, rejected cartridges which are reworked or repaired in accordance with government approved procedures shall be reinspected in 4 angular orientations approximately 90 degrees apart. Failure of the cartridge to be inserted or seated in any orientation shall be cause for rejection of the cartridge.

4.4.3.6 Ballistic tests. (See 4.5.6.)

4.4.3.6.1 Test plans. Summarized in Table II, the following tests shall be performed at a proving ground in accordance with established acceptance test procedures. Initial production acceptance shall consist of the 70° new and worn tube phases until three consecutive lots are accepted with respect to flight and safety. Thereafter, regular production acceptance shall consist of only the 70° new tube phase. Failure of the first article or production samples to comply with requirements of this specification shall result in rejection of the lot.

4.4.3.6.2 Muzzle velocity. (See 3.7.1) The sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code letter G, and an AQL of 1.5%.

4.4.3.6.3 Target impact dispersion. (See 3.7.2, 4.5.6.1.2) Sample cartridges used in the test for muzzle velocity shall be observed for target impact dispersion. If the pooled P.E. of the sample (calculated in accordance with 6.12) multiplied by the appropriate factor of Table III exceeds the requirement, the lot shall be rejected (see 6.12).

TABLE III

Factors for computing acceptability of P.E.

Sample Size	Degrees of Freedom	Factor
9	7	.62
10	8	.63
14	11	.67
15	12	.68

4.4.3.6.4 Tracer functioning. (See 3.7.3) All sample cartridges shall be observed for tracer functioning. If two or more cartridges in any single phase fail to comply with requirements, the lot shall be rejected.

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4.4.3.6.5 Safety. (See 3.7.4) All sample cartridges shall be observed for hangfires or breakup of metal parts. If any cartridge fails to comply with requirements, the lot shall be rejected.

4.4.3.6.6 Flight. (See 3.7.5) If the number of cartridges exhibiting erratic flight exceeds the values listed in Table IV, the lot shall be rejected.

TABLE IV

Erratic flight acceptance criteria

Temperature (^{o}F)	Tube Wear	Quantity	Accept	Reject
+70	New	15	0	1
+70	Worn	15	1	2
+125	Mid-Life	15	1	2
+125	Worn	15		
-25	Mid-Life	15	1	2
-25	Worn	15		

Note: The accept/reject numbers for +125°F and -25°F apply for the total quantity of 30 for each temperature.

4.4.3.6.7 Spin rate. Spin rate data shall be collected for the following tests only:

a. First article test.

b. Production lot acceptance tests, until five successive lots have passed.

c. Retest conducted after failure of lot acceptance test.

d. Production lot acceptance tests for reworked lots.

e. Special tests when specifically directed.

4.4.4 Misfire. (See 3.7.6) A misfire attributable to the ammunition shall be cause for referral of the lot.

4.4.5 Inspection equipment. See 6.3 and MIL-A-48078.

4.4.5.1 Government rights to documentation. Inspection equipment drawings may be used by DOD activities whenever inspection equipment drawings are needed.

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4.5 Methods of inspection.

4.5.1 Cartridge case crimp (bullet pull). Cartridges shall be tested in an appropriate tensile testing machine to determine the security of the crimp of the cartridge case to the projectile. The load shall be applied in an axial direction with the rate of travel of the testing machine head from 0.100 to 0.125 inch per minute. Projectiles shall be completely separated from cartridge cases and inspected for damage. Damaged projectiles shall not be recased. Discovery of significant obturator and rotating band damage caused by faulty case assembly operation shall cause rejection of the lot.

4.5.2 Electrical resistance of primer circuit. Electrical resistance shall be measured between the primer electrode and the cartridge case with a maximum current of 20 milliamperes flowing through the primer.

4.5.3 Security of sub-projectile in sabot. The projectile shall be held in a fixture by the sabot and a torque of 50 ± 5 in. lbs. applied to the sub-projectile in a counterclockwise direction when facing the nose. No rearward force shall be applied to the sub-projectile during application of torque. Observation shall be made of rotation of the sub-projectile.

4.5.4 Rotating band looseness and seating. The rotating band of each projectile shall be subjected to a manually applied rotational force in a clockwise direction (as viewed from the rear of the projectile). Any assembly which exhibits rotation of the rotating band shall be removed from the lot. A projectile may be resubmitted for inspection not less than two hours after being retightened. In addition, each resubmitted projectile shall be inspected for presence of a gap between the rotating band and sabot with a 0.002 inch feeler gage. Any assembly with a gap shall be removed from the lot. Feeler gage must not enter more than one-fourth of the projectile circumference in total.

4.5.5 Profile and alignment check. The profile and alignment of each crimped cartridge shall be verified by a gage corresponding to dwg. 8828098. The gage will be operated in a horizontal position (0 degrees ± 2.5 degrees). The base of the cartridge shall be positioned within one inch of the push-arm of the gage. Insertion shall not exceed a peak force of 60 pounds (27.2 kg) or a peak velocity of 2 ft/sec (61 cm/sec). No manual assistance (i.e., rotation, agitation, etc.) is permitted during inspection, nor shall the gage be utilized as an alignment tool.

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4.5.6 Ballistic tests.

4.5.6.1 Test methods.

4.5.6.1.1. General. Sample cartridges shall be fired from a tank-mounted M68 gun and observed for conformance to all requirements. The following applies to all temp/tube test phases.

4.5.6.1.1.1 Camera coverage. Side smear cameras shall be positioned 22 ± 0.5 ft. (6.71 ± 0.15 meters) and 105 ± 5 ft. (32.00 ± 1.52 meters) from the muzzle.

4.5.6.1.1.2 Target specifications. The target shall have minimum dimensions of 20 ft. x 20 ft. (6.1 meters x 6.1 meters) and and shall be positioned at a range of 2000 meters. A new target shall be used for each test series.

4.5.6.1.1.3 Temperature conditioning. Ammunition shall be conditioned for approximately 24 hours to the following temperatures, as required, until all propellant has stabilized:

-25°F	=	-25°F + 5°F
70°F	=	70°F ± 2.5°F
125°F	=	125°F - 5°F

4.5.6.1.1.4 Gun zeroing - boresight and superelevation. Gun superelevation is the difference between target center boresight and test phase elevations. Superelevation shall be measured for all target impact dispersion test phases. This requires that the gun be zeroed at the beginning, during any breaks in excess of 30 minutes and at the end of TID test phases, using a conventional collimated M26 Pye Watson boresight or equivalent.

The procedure is to first record elevation and azimuth which aligns the boresight zero on target center, and then to record elevation and azimuth for all test and calibration phases. Then determine superelevation by computing the difference. Superelevation data shall be entered on the ballistic test report.

4.5.6.1.1.5 Gun tube backlash removal. After loading and locking a round in the chamber, the gun tube shall be traversed off the target center, dropped down below the target center, traversed to the opposite side of the target center, elevated to the target center elevation, and traversed onto the target center. The distance traversed and dropped should be sufficient to remove the backlash from the system. If the sight is above or beyond the target center, do not reverse direction of gun tube movement, but repeat the sequence.

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4.5.6.1.1.6 Spotter round firing. Spotter rounds will be used to assure the weapon is zeroed on target center for all test phases. A minimum of three consecutive target hits within one mil of target center is required, and the three hits shall be obtained using the same elevation and deflection settings. After the target has been spotted for a shot group, every effort should be made to keep the elevation and deflection of the weapon the same for that shot group.

4.5.6.1.1.7 Test round firing. Test samples shall be fired in five round groups with no less than three nor more than five minutes between rounds. If a lapse exceeding thirty minutes occurs between group firings or if there is an indication of a trend away from target center, the gun will be re-zeroed.

4.5.6.1.1.8 Tracer functioning. Cartridges fired during all tests shall be observed for tracer functioning to the target. The distance over which the tracer burns shall be recorded to the nearest estimated 500 meters (i.e. failure, 500, 1000, 1500 meters).

4.5.6.1.1.9 Flight. The observer will be located at an elevated point behind the firing vehicle with an unobstructed view of the line of flight to the target and shall document projectile flight. Any abnormality of flight will be described (corkscrewing, tumbling, yaw, etc.,) as well as target impact information (hit target, short by 100 meters, left of target, etc.). Target miss coordinates will be estimated and recorded for all test phases along with pertinent flight observations.

4.5.6.1.1.10 Wind velocity. The velocity/target impact-dispersion portion of this ballistic test shall not be conducted when crosswinds exceed 10 knots or when gusts exceed 5 knots or when peak surface winds (crosswinds plus gusts) exceed 10 knots during the firing of any shot group. Wind data is not required for temperature extreme test samples.

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4.5.6.1.2 70°F New tube phase - (MV - TID).

4.5.6.1.2.1 General. Data generated in this phase will be used primarily for determination of acceptance from a target impact dispersion (TID)/muzzle velocity (MV) standpoint and will require the firing of calibration rounds to establish a velocity correction factor. This factor will then be applied to the actual test round velocities to obtain the corrected velocities on which the decision to accept or reject will be based. Cartridge performance used as acceptance criteria for this phase are: muzzle velocity, target impact dispersion, tracer functioning, safety malfunctions, misfires and flight characteristics.

Performance data taken for information only: spin rate (see 4.4.3.5.7 for spin rate schedule).

4.5.6.1.2.2 Test method.

a. Temperature condition calibration and test rounds at 70°F.

b. Fire calibrations and test rounds in a new gun tube the following sequence:

5 calibration rounds
15 test rounds
5 calibration rounds

c. Observe muzzle velocity, target impact dispersion, tracer function, flight, safety malfunctions, misfires and spin rate.

d. Measure target impact coordinates to the nearest 1/2 inch.

e. Compute the pooled probable error.

4.5.6.1.3 -25°F, 70°F, and 125°F Mid-life and worn tube phases.

4.5.6.1.3.1 General. The -25°F, 70°F, and 125°F mid-life and worn tube phases apply only to first article and provide additional flight characteristics acceptance data to supplement muzzle velocity and target impact dispersion obtained in the 70°F new tube phase.

Cartridge performance used as acceptance criteria for these phases are: tracer functioning, safety malfunctions, misfires and flight characteristics.

Performance data taken for information only: muzzle velocity, target impact dispersion, and spin rate. (See 4.4.3.5.7 for spin rate schedule.)

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4.5.6.1.3.2 Test method. Same as for 70°F new tube phase, except calibration rounds are not required.

4.5.6.2 Retest. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional cartridges, as required, shall be tested. Results of retests shall be referred to the contracting officer for disposition.

4.5.6.2.1 Target impact dispersion retest. A lot shall be retested whenever the lot fails to meet the target impact dispersion requirements in the presence of one or more of the following conditions:

a. The wind velocity exceeds 10 knots (18.5 km/hr) during firing of any test group.

b. The cadence between any two consecutive rounds in a test group exceeds 5 minutes or not exceeds three minutes.

c. A change was made in the azimuth or elevation of the weapon within a test group.

d. Both the test samples and the calibration samples fail to meet the target impact dispersion requirements.

e. If one of the test samples missed the target, the lot shall be retested if the pooled P.E. of the remaining samples multiplied by the appropriate factor meets the requirements. Two or more target misses by the test samples shall reject the lot. Target miss coordinates shall be estimated and recorded along with pertinent flight observations.

4.5.6.2.2 Velocity retest. A lot shall be retested whenever the lot fails to meet the velocity requirements in the presence of one or more of the following conditions:

a. The muzzle velocity correction factor computed from the average muzzle velocity of the calibration samples exceeds 25 fps (7.62 mps).

b. Two or more velocities are lost.

c. The wind velocity exceeds 10 knots (18.5 km/hr) during the firing.

d. Both the test samples and the calibration samples fail to meet the velocity requirements.

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5. PACKAGING

5.1 Preservation and packaging. The cartridge shall be preserved and packaged in accordance with dwg. 9278417.

5.2 Packing. The cartridge shall be preserved and packaged in accordance with dwg. 9278419 or 9341656.

5.3 Marking. Containers shall be marked in accordance with dwgs. 9278417 and 9278419 or 9341656.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Specifications conforming to the requirements of this standard are intended for use as military standardization documents and are listed in the DODISS. The general format described should also be considered for use in developing purchase descriptions and other non-DODISS procurement specifications, especially those which may be converted to a military specification at a later date.

6.2 Acquisition requirements. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6.3 Erratic flight definition. Erratic flight is any abnormality in the subprojectile flight path, usually displayed by corkscrewing, tumbling or undulation.

6.4 Cost of primer check test. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of check tests. The tests shall be conducted at Government expense without cost to the contractor who loaded the assemblies or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

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6.5 Submission of test data and ammunition data cards. In addition to the normal distribution of records, when the cartridge is procured by the US Army Materiel Readiness Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following:

Commander
U.S. Army Armament Research, Development and Engineering Center
ATTN: SMCAR-CCH-V and AMSMC-QAT-A (D)
Picatinny Arsenal, NJ 07806-5000

6.6 International standardization. Certain provisions of this specification are the subject of international standardization agreement ABCA-ARMY-STD-147. When amendment, revision or cancellation of this specification is proposed, the Departmental Custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.7 Ballistics test summary. See Table II.

6.8 Selection of ballistic samples. Whenever two projectile lots are used in a cartridge lot, the ballistic test sample shall be selected proportionally to represent the quantity of each projectile lot used.

6.9 Spotters. The spotter rounds used during any ballistic test shall ballistically match the test rounds. The weight of the assembled spotter round shall match the weight of the round to be tested. Five spotter rounds shall be available at the proving ground during each production lot acceptance test. Fifteen spotters shall be available at the proving ground during the first article ballistic test. Spotter rounds used shall be of the same configuration as the test rounds.

6.10 Firing record information. Whenever two projectile lots are used in a cartridge lot, test samples shall be identified as to projectile lot used in the firing record. Target impact plot, meteorological data and stargage measurements shall also be included in all firing records. The firing records shall include the serial numbers of all weapon and fire control components, velocity reference/calibration ammunition component data where used, test ammunition component data and weapon to target azimuth. Each round fired during the test shall be identified in the sequence of firing. For each round fired, the record shall contain deflection, elevation, time of firing, muzzle velocity, tracer functioning, coordinates of impact, and flight characteristics. Chamber pressures shall be recorded for each first article round fired, and for each reference round fired during production tests.

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6.11 Gun tube wear limits. For the purpose of this test a new tube is defined as having no greater than .015 inches primary land wear and a midlife tube as having no greater than .030 inches primary land wear. A worn tube is defined as follows:

Primary Land Wear Limits	Secondary Land Wear Limits	Secondary Groove Wear Limits
Greater than 0.030 in. and less than .056 in.	Between 70% and 40% of primary land wear	Between 80% and 190% of primary land wear

6.12 Method for calculating pooled probable error (P.E.). The pooled P.E. shall be defined as the square root of the weighted average of the squares of the individual P.E.s using the degrees of freedom as weights. The individual P.E. results of the three groups of five samples each shall be pooled according to the following method:

$$\text{PE pooled (mils)} = \sqrt{\frac{\Sigma\,(N_i - 1)\,PE_i^2}{\Sigma\,(N_i - 1)}}$$

Where: N_i = Sample size of each group (5 rounds)
PE_i = PE of each 5 round group

6.13 Probable error factor. The factors listed in paragraph 4.4.3.5.3 are taken from Table A-21 of AMC Pamphlet 706-114 using the 99 percent upper confidence limits ($A_{.01}$) and $\Sigma\,(N_i - 1)$ degrees of freedom. To define $\Sigma\,(N_i - 1)$ see 6.12.

6.14 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as ARDC, ARRADCOM, Edgewood, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.15 Subject term (key word) listing.

Ammunition
Cartridge
Container
Propellant
Projectile
Ballistic test
Target impact dispersion
First article inspection
Primers
Spotters

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6.16 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-A729)

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